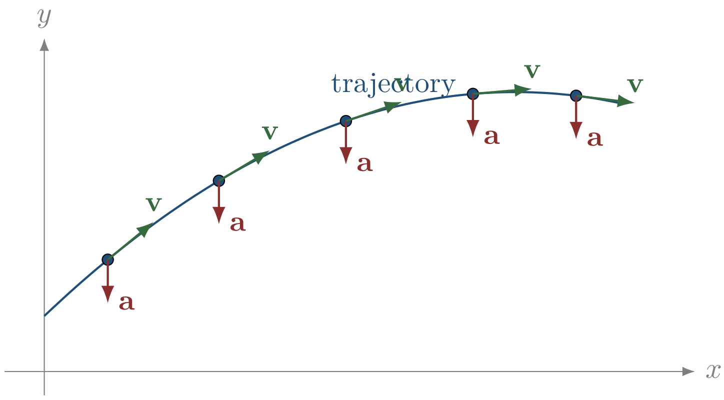
Equations of Motion

A Narrative Lecture on Pure Kinematics: Position, Velocity, Acceleration, Differentiation, Integration, Initial Conditions, and Trajectories



motion is a curve plus a clock

Lecture notes for students who must learn to read functions, derivatives, integrals, and graphs as a language for describing motion.

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Contents

How to read these notes

These notes are about kinematics. Kinematics is the part of mechanics in which we describe motion before we ask what force causes it. This distinction matters. In dynamics we ask why motion changes. In kinematics we ask how to describe the motion once it is happening. We define position, velocity, and acceleration. We learn how to read them from functions and graphs. We learn how differentiation and integration move us between these quantities. We learn why initial conditions are not small technical details, but essential information that selects one motion from a whole family of possible motions.

The main purpose of this document is not to provide a list of formulas. The main purpose is to teach a way of thinking. A function such as

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2$$

is not just a box into which numbers are inserted. It is a compact story about motion. It says where the object was at the initial moment, how fast it initially moved, and how its velocity changed with time. If we differentiate it, we do not perform an empty algebraic ritual. We ask: how does position change at this instant? If we integrate acceleration, we do not merely reverse differentiation. We reconstruct motion from information about change, and we must supply the missing constants by giving initial conditions.

Throughout the text, pay attention to the words surrounding the equations. In a well-written physical solution, equations do not stand alone. They are introduced by physical meaning and followed by interpretation. A calculation should have a rhythm:

$$\text{physical question} \longrightarrow \text{mathematical translation} \longrightarrow \text{calculation} \longrightarrow \text{physical reading.}$$

The calculation is the middle of the story, not the whole story.

The central message

Kinematics is the grammar of motion. Position, velocity, and acceleration are not separate formulas. They are related descriptions of the same motion, connected by differentiation and integration.

1 What kinematics is actually about

Before any derivative appears, we should understand what we are trying to describe. Motion is not simply the fact that an object is somewhere. Motion means that the object's position is related to time. A photograph can show the position of a car. It cannot by itself show whether the car is moving, how fast it is moving, or whether it is speeding up. To speak about motion we need a clock.

In elementary mechanics we usually model the moving body as a particle. This does not mean that the object is literally a point. It means that, for the question we are asking, its size, shape, rotation, and internal structure are not important. A car on a highway may be treated as a particle if we only care about where it is along the road. The same car cannot be treated as a particle if we study how its wheels rotate or how it turns around its own centre of mass.

A kinematic description therefore begins with three choices:

1. We choose the object whose motion is being described.
2. We choose a reference frame and coordinate system.
3. We choose a time variable and decide what moment is called $t = 0$.

Only after these choices does an expression such as $x(t)$ or $\mathbf{r}(t)$ have a clear meaning.

1.1 Position as information, not just a number

In one dimension the position may be described by a single coordinate $x(t)$. In two dimensions we need two coordinates, for example

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}.$$

In three dimensions we write

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

The vector notation is not decorative. It reminds us that position has direction relative to the chosen origin. The coordinate functions $x(t)$, $y(t)$, and $z(t)$ are components of one object: the position vector.

When students first see these expressions, they often treat them as formal symbols. It is better to read them as a precise sentence. The expression

$$\mathbf{r}(t) = (3t)\mathbf{i} + (2 + 4t - t^2)\mathbf{j}$$

says the following: after t seconds, the horizontal coordinate is $3t$, while the vertical coordinate is $2 + 4t - t^2$. The object is not merely "on a parabola". It moves along a path according to a specific time schedule.

Before calculating

When you see a position function, first read it in words. Ask what each component says about how the position changes with time. Only after that should you start differentiating or substituting numbers.

1.2 Motion and trajectory are not the same thing

A trajectory is the geometric path traced by the object. Motion is the trajectory together with the information about time. This distinction is simple, but it prevents many mistakes.

Consider the two motions

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j}, \qquad \mathbf{r}_2(t) = t^2\mathbf{i} + t^4\mathbf{j}.$$

Both satisfy the geometric relation $y = x^2$ for $x \geq 0$. Therefore, the two objects move along the same curve in the plane. But they do not move in the same way. In the first motion, $x = t$. In the second, $x = t^2$. The second object moves slowly near $t = 0$ and then faster later. The curve is the same, but the timing is different.

This is why it is dangerous to look only at the shape of the path. A drawing of the path tells us where the object may be. It does not fully tell us how it moves.

Same path, different motion

Let

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j}, \qquad \mathbf{r}_2(t) = 2t\mathbf{i} + 4t^2\mathbf{j}.$$

Both motions lie on the same parabola $y = x^2$ after eliminating time? For $\mathbf{r}_2$, $x = 2t$, so $t = x/2$, and $y = 4t^2 = x^2$. Yes, the path is the same. But $\mathbf{r}_2$ reaches each point earlier. The trajectory is geometric; motion is geometric and temporal.

2 Average velocity and instantaneous velocity

Velocity begins as a comparison between two positions at two times. Suppose that at time t_1 an object is at position $x(t_1)$, and at time t_2 it is at position $x(t_2)$. The displacement is

$$\Delta x = x(t_2) - x(t_1),$$

and the elapsed time is

$$\Delta t = t_2 - t_1.$$

The average velocity over this interval is

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}.$$

This expression is not yet a derivative. It is a ratio of two finite changes. It answers the question: over the whole time interval, how much position changed per unit time.

The word “average” is important. A car may drive slowly at first and quickly later. The average velocity over the whole trip does not tell us the velocity at every moment. It compresses the trip into one number.

2.1 The idea of instantaneous velocity

To describe motion more precisely, we shrink the time interval. We ask what the average velocity becomes when the second time is very close to the first. This leads to the derivative:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}.$$

The derivative is the instantaneous velocity. It is the rate at which position is changing at that moment.

The symbol $\mathrm{d}x/\mathrm{d}t$ should not be read as a mysterious decoration. It says: compare a very small change in position with the corresponding very small change in time. The derivative is the mathematical object that makes this idea precise.

Velocity as a local description

Average velocity describes change over an interval. Instantaneous velocity describes the local rate of change at one moment. The derivative is the limit that turns interval information into local information.

2.2 Velocity in several dimensions

For vector motion the same idea applies, but now position is a vector:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

The velocity vector is

$$\mathbf{v}(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\mathbf{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\mathbf{j} + \frac{\mathrm{d}z}{\mathrm{d}t}\mathbf{k}.$$

So the components of velocity are the derivatives of the coordinate functions:

$$v_x = x'(t), \qquad v_y = y'(t), \qquad v_z = z'(t).$$

This is not a new rule invented for vectors. It is the same derivative applied component by component.

2.3 Speed is not velocity

Velocity is a vector. Speed is its magnitude. In one dimension, students often blur this distinction because a positive velocity and a speed may look the same numerically. In two or three dimensions, the distinction becomes unavoidable.

For a velocity vector

$$\mathbf{v}(t) = v_x(t)\mathbf{i} + v_y(t)\mathbf{j} + v_z(t)\mathbf{k},$$

the speed is

$$|\mathbf{v}(t)| = \sqrt{v_x(t)^2 + v_y(t)^2 + v_z(t)^2}.$$

The velocity tells us how fast and in what direction the object moves. The speed tells us only how fast.

Common mistake

A negative velocity in one-dimensional motion does not mean a negative speed. It means motion in the negative direction of the chosen axis. Speed is never negative.

3 Acceleration: how velocity changes

Acceleration is often introduced too quickly as “the second derivative of position”. That statement is correct, but it hides the meaning. Acceleration is the rate of change of velocity. If velocity changes, the object accelerates. The change may be a change in speed, a change in direction, or both.

In one dimension,

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}.$$

In vector form,

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2 \mathbf{r}}{\mathrm{d}t^2}.$$

Again, this is not merely a formal operation. The first derivative tells us how position changes. The second derivative tells us how the change of position changes.

3.1 Acceleration without increasing speed

A very important idea is that acceleration does not always mean “getting faster”. If an object moves around a circle at constant speed, its velocity direction changes continuously. Therefore, the velocity vector changes, and the object has acceleration even though its speed remains constant.

This is one reason vector thinking is necessary. In everyday language, “accelerate” often means increase speed. In physics, acceleration means change of velocity. Since velocity includes direction, a direction change is acceleration.

Acceleration has two possible effects

Acceleration may change the magnitude of velocity, the direction of velocity, or both. The question is not only whether the object is getting faster. The question is whether the velocity vector is changing.

3.2 Reading signs in one dimension

In one-dimensional motion, sign matters. If $v > 0$, the object moves in the positive direction. If $v < 0$, it moves in the negative direction. If $a > 0$, velocity is increasing. If $a < 0$, velocity is decreasing.

But this does not mean that negative acceleration always slows the object down. Suppose $v < 0$ and $a < 0$. Then the velocity becomes more negative. The object moves faster in the negative direction, so its speed increases. To decide whether the speed increases, compare the signs of v and a:

- if v and a have the same sign, the speed increases;
- if v and a have opposite signs, the speed decreases.

This is a simple example of why mechanical interpretation cannot be replaced by blind algebra.

3.3 A complete reading of a simple function

Let

$$x(t) = 2 + 3t - t^2.$$

The velocity is

$$v(t) = x'(t) = 3 - 2t,$$

and the acceleration is

$$a(t) = v'(t) = -2.$$

Now read this in words. The object starts at $x(0) = 2$. Its initial velocity is $v(0) = 3$, so initially it moves in the positive direction. Its acceleration is constant and negative, so its velocity decreases linearly. At

$$3 - 2t = 0$$

we get

$$t = \frac{3}{2}.$$

At this time the object momentarily stops. After that, the velocity becomes negative, so the object moves back in the negative direction.

The calculation is short, but the interpretation is rich. We learned initial position, initial velocity, acceleration, the stopping time, and the qualitative story of the motion.

4 Differentiation as a machine for extracting motion information

If position is known as a function of time, differentiation extracts velocity and acceleration. This is the most direct route through kinematics:

$$\boxed{\mathbf{r}(t) \longrightarrow \mathbf{v}(t) = \mathbf{r}'(t) \longrightarrow \mathbf{a}(t) = \mathbf{r}''(t).}$$

But it is important to avoid treating this as a mechanical ritual. We differentiate because we are asking about change.

4.1 Example: polynomial position

Suppose

$$x(t) = 1 + 4t - 3t^2 + t^3.$$

The velocity is

$$v(t) = 4 - 6t + 3t^2,$$

and the acceleration is

$$a(t) = -6 + 6t.$$

The first derivative is a quadratic function. The second derivative is a linear function. That means the velocity changes in a non-uniform way, and the acceleration itself changes with time.

At $t = 0$ we have

$$x(0) = 1, \qquad v(0) = 4, \qquad a(0) = -6.$$

At $t = 1$ we have

$$x(1) = 3, \qquad v(1) = 1, \qquad a(1) = 0.$$

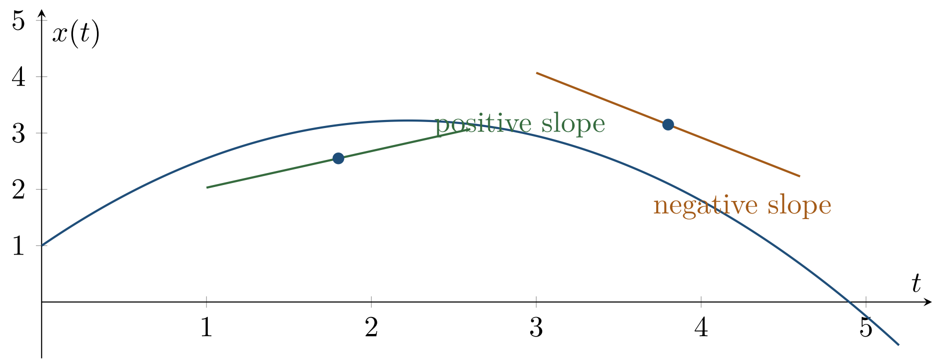
At $t = 2$ we have

$$x(2) = 5, \qquad v(2) = 4, \qquad a(2) = 6.$$

These values should not be read as isolated numbers. They show a story: the object starts with positive velocity, slows down because acceleration is initially negative, reaches a moment where acceleration is zero, and later acceleration becomes positive.

4.2 The derivative as slope of a graph

The derivative has a geometric meaning. On a graph of x versus t, the velocity is the slope of the tangent line. If the graph is steep and increasing, velocity is large and positive. If the graph is flat, velocity is zero. If the graph decreases, velocity is negative.



This picture is not an extra illustration. It is another way of saying the same thing as the derivative. A graph, a derivative, and a verbal description are three languages for the same relationship.

4.3 Local approximation

The derivative also tells us how to approximate the function locally. If $x(t)$ is known at time t_0, then for a small time change Δt,

$$x(t_0 + \Delta t) \approx x(t_0) + v(t_0)\Delta t.$$

This means that over a very short time interval, the motion looks almost uniform. The instantaneous velocity is the velocity of the best local straight-line approximation.

If we include acceleration, we get a better approximation:

$$x(t_0 + \Delta t) \approx x(t_0) + v(t_0)\Delta t + \frac{1}{2}a(t_0)(\Delta t)^2.$$

This formula is often introduced as a memorized equation. But here we can read it as a local description: position after a short time is initial position plus the contribution from initial velocity plus a correction due to acceleration.

5 Integration: reconstructing motion from change

Differentiation moves from position to velocity to acceleration. Integration moves in the opposite direction:

$$\boxed{\mathbf{a}(t) \longrightarrow \mathbf{v}(t) \longrightarrow \mathbf{r}(t).}$$

This route is deeper than it first appears. When we integrate, we reconstruct a quantity from its rate of change. But a rate of change alone does not determine the original quantity uniquely. We need constants of integration. Physically, those constants are initial conditions.

5.1 From acceleration to velocity

If acceleration is known, then velocity changes according to

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(s)\,\mathrm{d}s.$$

This expression says: the velocity at time t equals the initial velocity plus the accumulated acceleration from t_0 to t.

In one dimension,

$$v(t) = v_0 + \int_{t_0}^{t} a(s)\,\mathrm{d}s.$$

If a is constant and $t_0 = 0$, then

$$v(t) = v_0 + at.$$

This is not a separate magic formula. It is the integral of constant acceleration.

5.2 From velocity to position

Similarly,

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(s)\,\mathrm{d}s.$$

In one dimension,

$$x(t) = x_0 + \int_{t_0}^{t} v(s)\,\mathrm{d}s.$$

If $v(t) = v_0 + at$ and $t_0 = 0$, then

$$x(t) = x_0 + \int_0^t (v_0 + as)\,\mathrm{d}s = x_0 + v_0 t + \frac{1}{2}at^2.$$

This derivation is important because it shows where the familiar expression comes from. It is not a primitive fact. It is the result of integrating velocity, which itself was obtained by integrating acceleration.

Constants are physical information

When you integrate acceleration, you must know initial velocity. When you integrate velocity, you must know initial position. The constants of integration are not mathematical noise. They are the physical data needed to select the actual motion.

5.3 Area under a graph

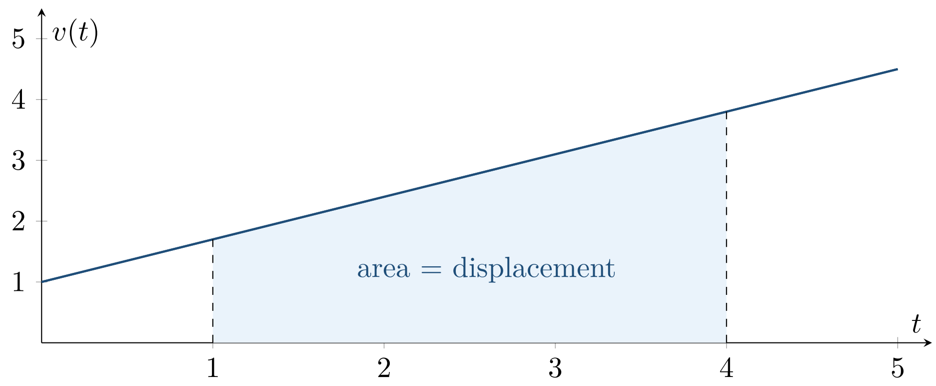
Integration also has a graphical interpretation. The area under a velocity-time graph gives displacement:

$$\Delta x = \int_{t_1}^{t_2} v(t)\,\mathrm{d}t.$$

The area under an acceleration-time graph gives change in velocity:

$$\Delta v = \int_{t_1}^{t_2} a(t)\,\mathrm{d}t.$$

This is why graphs are not decorative pictures. A graph can contain quantitative information. Slope and area are two fundamental ways of reading motion.



6 Initial conditions and families of motion

One of the most important ideas in kinematics is that an equation involving derivatives usually describes a family of possible motions. Initial conditions choose one member of the family.

Suppose we know that acceleration is constant:

$$a(t) = a.$$

Integrating once gives

$$v(t) = at + C_1.$$

The constant C_1 is the velocity at $t = 0$ if the initial time is chosen as zero. We therefore write

$$v(t) = v_0 + at.$$

Integrating again gives

$$x(t) = \frac{1}{2}at^2 + v_0 t + C_2.$$

The constant C_2 is the position at $t = 0$, so

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

Notice what happened. The acceleration function alone did not give one motion. It gave all motions with that acceleration. Different choices of x_0 and v_0 produce different motions.

6.1 Same acceleration, different motion

Take $a = -g$. This describes constant downward acceleration. But the following are different motions:

$$y_1(t) = 10 - \frac{1}{2}gt^2,$$

$$y_2(t) = 10 + 5t - \frac{1}{2}gt^2,$$

$$y_3(t) = 2 - 3t - \frac{1}{2}gt^2.$$

They all have the same acceleration, because

$$y_1''(t) = y_2''(t) = y_3''(t) = -g.$$

But they describe different initial positions and initial velocities. The first is dropped from height 10. The second is thrown upward from height 10 with initial velocity 5. The third starts at height 2 and already moves downward with velocity -3.

Common mistake

Do not confuse knowing acceleration with knowing the whole motion. To reconstruct position from acceleration, you also need initial velocity and initial position.

6.2 Changing the zero of time

Initial conditions depend on the chosen time origin. If we choose $t = 0$ at launch, then x_0 and v_0 are launch data. If we choose $t = 0$ at some later instant, the numerical values of the initial conditions change. The physical motion does not change, but our description of it does.

For example, suppose

$$x(t) = 2 + 3t + t^2.$$

If we choose a new time variable $\tau = t - 1$, then $t = \tau + 1$, and

$$x = 2 + 3(\tau + 1) + (\tau + 1)^2 = 6 + 5\tau + \tau^2.$$

The same physical function has a different appearance in the new time coordinate. At $\tau = 0$, which corresponds to $t = 1$, the position is 6 and the velocity is 5. This does not contradict the original expression. It only reflects a different choice of clock origin.

7 Constructing equations of motion from words

Students often ask for “the formula”. In kinematics, a better question is: what information has been given, what quantity is changing, and what conditions must be satisfied? The equation of motion should be constructed from the situation.

7.1 Uniform motion

Suppose an object moves in a straight line with constant velocity. The phrase “constant velocity” means

$$v(t) = v_0.$$

Since velocity is the derivative of position,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_0.$$

Integrating gives

$$x(t) = x_0 + v_0 t.$$

The equation says: start at x_0, then add the same displacement v_0 during each unit of time.

If $v_0 > 0$, the object moves in the positive direction. If $v_0 < 0$, it moves in the negative direction. If $v_0 = 0$, the object remains at the same position.

7.2 Uniformly accelerated motion

Suppose the acceleration is constant. The physical sentence is

$$a(t) = a.$$

Then

$$v(t) = v_0 + at,$$

and

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

This is the standard uniformly accelerated motion. But the important point is not the final expression itself. The important point is the chain of meaning:

$$\text{constant acceleration} \Rightarrow \text{linear velocity} \Rightarrow \text{quadratic position.}$$

So when you see a quadratic position function, you should suspect constant acceleration. When you see a linear velocity function, you should also suspect constant acceleration.

7.3 Motion with time-dependent acceleration

Now suppose acceleration is not constant. For example,

$$a(t) = kt.$$

This says acceleration grows linearly with time. Then

$$v(t) = v_0 + \int_0^t ks\,\mathrm{d}s = v_0 + \frac{1}{2}kt^2,$$

and

$$x(t) = x_0 + \int_0^t \left(v_0 + \frac{1}{2}ks^2 \right) \mathrm{d}s = x_0 + v_0 t + \frac{1}{6}kt^3.$$

The pattern changes. Linear acceleration gives quadratic velocity and cubic position. Again, the equation of motion is built by integration, not found by memorization.

Building motion from verbal data

Translate the sentence into a statement about x, v, or a. Then use differentiation or integration to move to the desired quantity. Finally, use initial conditions to determine constants.

8 The standard constant-acceleration relations

The constant-acceleration case is so common that it produces several familiar relations. These relations are useful, but they should be understood as consequences of a model, not as independent formulas.

Assume motion in one dimension with constant acceleration a and initial time $t = 0$. Then

$$v(t) = v_0 + at,$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

The displacement is

$$\Delta x = x - x_0 = v_0 t + \frac{1}{2}at^2.$$

8.1 Eliminating time

Sometimes time is not directly needed. We can eliminate it. From

$$v = v_0 + at$$

we get

$$t = \frac{v - v_0}{a}$$

if $a \neq 0$. Substitute this into the displacement expression, or use a cleaner derivation:

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}.$$

Therefore,

$$a\,\mathrm{d}x = v\,\mathrm{d}v.$$

Integrating from x_0 to x and from v_0 to v gives

$$a(x - x_0) = \frac{1}{2}(v^2 - v_0^2),$$

so

$$v^2 = v_0^2 + 2a(x - x_0).$$

This expression is not a new law. It is the constant-acceleration model rewritten without time.

8.2 Average velocity for constant acceleration

For constant acceleration, velocity changes linearly. The average of the initial and final velocities is therefore

$$v_{\text{avg}} = \frac{v_0 + v}{2}.$$

Then

$$\Delta x = v_{\text{avg}} t = \frac{v_0 + v}{2}t.$$

This relation is useful when time and velocities are known. But it depends on velocity being linear in time. If acceleration is not constant, this simple average formula is not generally true.

Do not overgeneralize

The equations $v = v_0 + at$, $x = x_0 + v_0 t + \frac{1}{2}at^2$, and $v^2 = v_0^2 + 2a\Delta x$ assume constant acceleration. They are not universal kinematics formulas.

9 Graphs as equations in visual form

A graph is not a decorative addition to a solution. A graph is another representation of the same motion. If students learn to read graphs, they gain physical understanding that formulas alone often hide.

9.1 Position-time graphs

On a position-time graph, the vertical coordinate is position and the horizontal coordinate is time. The slope of this graph is velocity:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}.$$

If the graph rises, velocity is positive. If the graph falls, velocity is negative. If the graph is horizontal, velocity is zero.

The curvature of the graph contains information about acceleration. If the graph bends upward, acceleration is positive. If it bends downward, acceleration is negative. More precisely, acceleration is the second derivative:

$$a(t) = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}.$$

9.2 Velocity-time graphs

On a velocity-time graph, the slope is acceleration:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t}.$$

The area under the graph gives displacement:

$$\Delta x = \int v(t)\,\mathrm{d}t.$$

The sign of this area matters. Area above the time axis contributes positive displacement. Area below the time axis contributes negative displacement.

9.3 Acceleration-time graphs

On an acceleration-time graph, the area gives change in velocity:

$$\Delta v = \int a(t)\,\mathrm{d}t.$$

A positive area increases velocity. A negative area decreases velocity. But again, decreasing velocity is not always the same as decreasing speed. The interpretation depends on the direction of motion.

Three graph questions

When reading any kinematic graph, ask:

1. What quantity is on the vertical axis?
2. What does the slope mean?
3. What does the area mean, if anything?

The answer changes depending on whether the graph is $x(t)$, $v(t)$, or $a(t)$.

9.4 A graph-reading example

Suppose a velocity-time graph starts at $v = 2$, rises linearly to $v = 6$ over four seconds, and then remains constant. During the first four seconds the acceleration is positive and constant. The displacement during this interval is the area of a trapezoid:

$$\Delta x = \frac{2 + 6}{2} \cdot 4 = 16.$$

After that, the velocity is constant, so acceleration is zero and displacement grows linearly. This example can be solved without first writing a position function. The graph already contains the necessary information.

10 Vector kinematics and component thinking

Most real motions do not occur on a single line. Vector kinematics allows us to describe motion in the plane or in space. The central object is the position vector

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

Velocity and acceleration are obtained by differentiating component by component:

$$\mathbf{v}(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k},$$

$$\mathbf{a}(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}.$$

This component structure is powerful because many motions are easier to understand by separating directions.

10.1 Independent components

If

$$x(t) = 2t, \qquad y(t) = 5 - t^2,$$

then horizontal and vertical motion have different character. Horizontally,

$$v_x = 2, \qquad a_x = 0.$$

Vertically,

$$v_y = -2t, \qquad a_y = -2.$$

So the object moves horizontally with constant velocity while accelerating downward. The full motion is a combination of these two component motions.

The vector description is

$$\mathbf{r}(t) = 2t\mathbf{i} + (5 - t^2)\mathbf{j},$$

$$\mathbf{v}(t) = 2\mathbf{i} - 2t\mathbf{j},$$

$$\mathbf{a}(t) = -2\mathbf{j}.$$

Notice how much information is contained here. The acceleration is constant and vertical. The velocity changes only in its vertical component. The path is curved because one component is linear in time and the other is quadratic.

10.2 Eliminating time to find the trajectory

The trajectory is found by eliminating t. From $x = 2t$ we get

$$t = \frac{x}{2}.$$

Substitute into $y = 5 - t^2$:

$$y = 5 - \left(\frac{x}{2}\right)^2 = 5 - \frac{x^2}{4}.$$

This is the geometric path. But remember: the path equation does not tell the full motion. The parametric equations tell the motion because they include time.

Parametric motion versus trajectory

The equations $x = x(t)$ and $y = y(t)$ describe motion. The relation $y = y(x)$ describes the path. The path is what is drawn in space; the motion includes how the path is traversed in time.

11 Drawing trajectories, velocity, and acceleration

A good drawing in kinematics is not a pretty picture. It is a compressed physical argument. It should show the path, selected positions, velocity directions, and acceleration directions. These arrows help us see whether our equations make sense.

11.1 Velocity is tangent to the trajectory

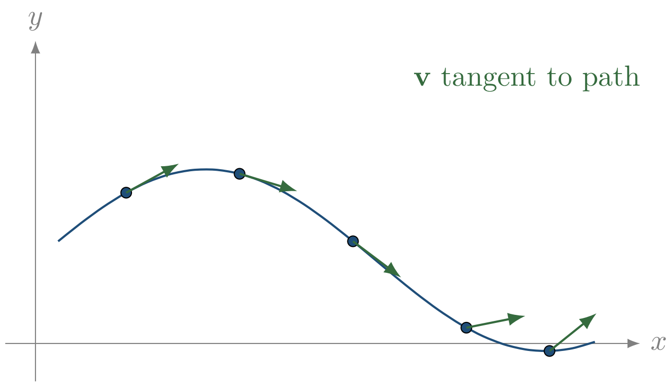
For any smooth trajectory, the velocity vector is tangent to the path. This follows directly from the definition

$$\mathbf{v}(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}.$$

Over a very short time interval Δt, the displacement is approximately

$$\Delta\mathbf{r} \approx \mathbf{v}(t)\Delta t.$$

Therefore, the velocity points in the direction of the small displacement along the curve.



11.2 Acceleration need not be tangent

Acceleration describes how velocity changes. It does not have to point along the path. In one-dimensional straight-line motion, acceleration is along the line because there is no other direction. In curved motion, acceleration may have a tangential component and a normal component.

The tangential component changes speed. The normal component changes direction. This idea becomes crucial in circular motion, but it is already a general kinematic idea.

11.3 A procedure for drawing motion

When drawing a trajectory with kinematic vectors, proceed as follows:

1. Draw the geometric path.
2. Mark several positions corresponding to increasing times.
3. Draw velocity arrows tangent to the path, pointing in the direction of motion.
4. Decide from the equations or from the change in velocity how acceleration should point.
5. Check whether the arrows agree with the algebraic signs and components.

This procedure forces a connection between formulas and physical meaning. If the drawing contradicts the equations, one of them has been misunderstood.

12 Tangential and normal acceleration

In curved motion, the velocity vector may change in two different ways. Its magnitude may change, and its direction may change. Kinematics separates these effects into tangential and normal acceleration.

Let the speed be

$$v = |\mathbf{v}|.$$

Let $\mathbf{T}$ be the unit tangent vector, pointing in the direction of motion. Then

$$\mathbf{v} = v\mathbf{T}.$$

Differentiate with respect to time:

$$\mathbf{a} = \frac{\mathrm{d}}{\mathrm{d}t}(v\mathbf{T}) = \frac{\mathrm{d}v}{\mathrm{d}t}\mathbf{T} + v\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t}.$$

The first term is tangential. It changes the speed. The second term is normal to the path. It changes the direction.

For motion along a curve with radius of curvature ρ, the normal part has magnitude

$$a_n = \frac{v^2}{\rho}.$$

Thus

$$\mathbf{a} = \frac{\mathrm{d}v}{\mathrm{d}t}\mathbf{T} + \frac{v^2}{\rho}\mathbf{N},$$

where $\mathbf{N}$ is the unit normal pointing toward the centre of curvature.

12.1 Why this matters

This formula teaches a very important lesson. Even if speed is constant, acceleration may not be zero. If the path is curved, the direction of velocity changes. Therefore, there is normal acceleration.

If the path is straight, the radius of curvature is infinite and the normal term disappears. Then acceleration can only be tangential. This is why one-dimensional motion hides part of the structure of kinematics.

Curved motion

Tangential acceleration changes speed. Normal acceleration changes direction. In curved motion, acceleration is not only about speeding up or slowing down.

12.2 Circular motion as a special case

For a circle, the radius of curvature is constant and equal to the circle radius R. Therefore,

$$a_n = \frac{v^2}{R}.$$

This expression belongs to kinematics. It does not yet tell us what force causes the acceleration. It only tells us what acceleration is required if the object moves along a circular path with speed v.

This prepares the bridge to dynamics, but the statement itself is purely kinematic: circular motion has inward acceleration because the velocity direction changes.

13 Polar coordinates as a language for motion

Cartesian coordinates are not always the most natural choice. If motion occurs around a point, polar coordinates may express the geometry more clearly. In the plane, a position can be written as

$$\mathbf{r}(t) = r(t)\mathbf{e}_r(t),$$

where $r(t)$ is the distance from the origin and $\mathbf{e}_r(t)$ is the unit vector pointing radially outward. The angular coordinate is $\theta(t)$.

The important complication is that the unit vectors themselves change with time. In polar coordinates,

$$\frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}t} = \dot{\theta}\mathbf{e}_\theta,$$
$$\frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}t} = -\dot{\theta}\mathbf{e}_r.$$

Therefore, differentiating position requires differentiating both the components and the basis vectors.

13.1 Velocity in polar coordinates

Start with

$$\mathbf{r} = r\mathbf{e}_r.$$

Differentiate:

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta.$$

This expression has a clear interpretation. The term $\dot{r}\mathbf{e}_r$ is radial velocity; it changes the distance from the origin. The term $r\dot{\theta}\mathbf{e}_\theta$ is transverse velocity; it changes the angle.

13.2 Acceleration in polar coordinates

Differentiate velocity:

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta.$$

After applying the product rule and the derivatives of the basis vectors, we obtain

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta.$$

This formula looks technical, but each term has a meaning. The term $-r\dot{\theta}^2\mathbf{e}_r$ is the inward acceleration associated with angular motion. The term $r\ddot{\theta}\mathbf{e}_\theta$ comes from changing angular speed. The term $2\dot{r}\dot{\theta}\mathbf{e}_\theta$ appears when radial motion and angular motion occur simultaneously.

Formula without context is dangerous

Polar-coordinate acceleration is not a formula to memorize blindly. It is the result of differentiating a vector expressed in a moving basis. The basis vectors change direction, and that is why extra terms appear.

13.3 Pure circular motion in polar coordinates

For motion on a circle of radius R,

$$r(t) = R, \qquad \dot{r} = 0, \qquad \ddot{r} = 0.$$

Then

$$\mathbf{v} = R\dot{\theta}\mathbf{e}_\theta,$$

and

$$\mathbf{a} = -R\dot{\theta}^2\mathbf{e}_r + R\ddot{\theta}\mathbf{e}_\theta.$$

If angular speed is constant, then $\ddot{\theta} = 0$, and

$$\mathbf{a} = -R\omega^2\mathbf{e}_r.$$

This is inward centripetal acceleration. Again, the result has been derived from kinematics, not assumed.

14 Equations of motion as differential equations

An equation of motion often appears as a differential equation. This means that the unknown function is related to its derivatives. Kinematics gives us the language to interpret such equations.

For example,

$$\ddot{x} = 0$$

means acceleration is zero. Integrating twice gives

$$x(t) = x_0 + v_0 t.$$

This is uniform motion.

The equation

$$\ddot{x} = a$$

means acceleration is constant. Integrating twice gives

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

The equation

$$\ddot{x} = -\omega^2 x$$

means acceleration is proportional to displacement and opposite in direction. This equation is often associated with dynamics and oscillators, but as a kinematic equation it says something about the shape of motion: the functions that satisfy it are sinusoidal,

$$x(t) = A\cos(\omega t) + B\sin(\omega t).$$

The constants A and B are determined by initial conditions.

14.1 Initial conditions for a second-order equation

A second-order equation involves the second derivative. To select one solution, we usually need two initial conditions:

$$x(0) = x_0, \qquad \dot{x}(0) = v_0.$$

This fits the physical interpretation. To know future one-dimensional motion under a given acceleration rule, we need both initial position and initial velocity.

For example, for

$$\ddot{x} = -\omega^2 x,$$

the general solution is

$$x(t) = A\cos(\omega t) + B\sin(\omega t).$$

The velocity is

$$v(t) = -A\omega\sin(\omega t) + B\omega\cos(\omega t).$$

At $t = 0$,

$$x(0) = A, \qquad v(0) = B\omega.$$

Thus

$$A = x_0, \qquad B = \frac{v_0}{\omega}.$$

So

$$x(t) = x_0\cos(\omega t) + \frac{v_0}{\omega}\sin(\omega t).$$

This is a complete solution because the constants have been connected to physical initial data.

15 The role of units and dimensions

Kinematic equations must be dimensionally consistent. This is not just a checking trick. Units carry meaning. Position has units of length, velocity has units of length per time, and acceleration has units of length per time squared:

$$[x] = \mathrm{m}, \qquad [v] = \mathrm{m/s}, \qquad [a] = \mathrm{m/s^2}.$$

When we write

$$x = x_0 + v_0 t + \frac{1}{2}at^2,$$

each term has units of length:

$$[x_0] = \mathrm{m}, \qquad [v_0 t] = (\mathrm{m/s})\mathrm{s} = \mathrm{m}, \qquad [at^2] = (\mathrm{m/s^2})\mathrm{s}^2 = \mathrm{m}.$$

This consistency is one reason the expression makes physical sense.

15.1 Dimension as a guard against mistakes

Suppose a student writes

$$x = x_0 + v_0 t + \frac{1}{2}at.$$

The last term has units

$$[at] = \mathrm{m/s},$$

which is velocity, not position. Therefore, the equation cannot be correct. Dimensional analysis has detected the error before any numbers were substituted.

Similarly, if the answer to a question asking for acceleration has units of metres, something has gone wrong. Units are part of the language of the solution.

Unit check

After deriving an expression, check the units before using numbers. This is not optional bookkeeping. It is a way to verify that the mathematical expression can represent the physical quantity you claim it represents.

15.2 Scaling and qualitative thinking

Units also help us think qualitatively. If acceleration is constant, the displacement caused by acceleration grows like t^2. This means doubling time multiplies that contribution by four. In contrast, displacement caused by initial velocity grows like t.

This difference matters. At short times, initial velocity may dominate. At longer times, acceleration may dominate. A formula becomes more meaningful when we read how its terms scale.

16 Worked example: reading a one-dimensional motion completely

Consider the motion

$$x(t) = 4 - 2t + t^2, \qquad t \geq 0.$$

We do not begin by asking what formula to use. The function already is the equation of motion. Our task is to read it.

16.1 Position, velocity, acceleration

Differentiate once:

$$v(t) = x'(t) = -2 + 2t.$$

Differentiate again:

$$a(t) = x''(t) = 2.$$

The acceleration is constant and positive. The initial position is

$$x(0) = 4,$$

and the initial velocity is

$$v(0) = -2.$$

So the object begins at $x = 4$, moves in the negative direction, but has positive acceleration.

16.2 Turning point

The object changes direction when velocity is zero:

$$-2 + 2t = 0.$$

Thus

$$t = 1.$$

At this time,

$$x(1) = 4 - 2 + 1 = 3.$$

So the object moves left until it reaches $x = 3$, stops momentarily, and then moves right.

16.3 Interpretation

The motion is a parabola on the x-versus-t graph. Because the coefficient of t^2 is positive, the graph opens upward. The minimum of the graph occurs at $t = 1$. This minimum is not merely a feature of a graph. It is the physical turning point.

What was actually done?

We used differentiation to find velocity and acceleration. We solved $v(t) = 0$ because a change of direction in one-dimensional motion occurs when velocity passes through zero. Then we substituted that time back into the position function to find where the turning point occurs.

This example shows the proper order of thought: identify the physical event, translate it into a mathematical condition, solve the condition, and interpret the result.

17 Worked example: reconstructing motion from acceleration

Suppose the acceleration of an object moving along a line is

$$a(t) = 6t.$$

At $t = 0$, the object has position

$$x(0) = 2$$

and velocity

$$v(0) = -1.$$

We want the position function $x(t)$.

17.1 First integration: acceleration to velocity

Acceleration is the derivative of velocity:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 6t.$$

Integrate from 0 to t:

$$v(t) - v(0) = \int_0^t 6s\,\mathrm{d}s.$$

Therefore,

$$v(t) - (-1) = 3t^2,$$

and

$$v(t) = -1 + 3t^2.$$

The initial condition has entered the calculation. Without it, we would know only that $v(t) = 3t^2 + C$.

17.2 Second integration: velocity to position

Now use

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v(t) = -1 + 3t^2.$$

Integrate:

$$x(t) - x(0) = \int_0^t (-1 + 3s^2)\,\mathrm{d}s.$$

So

$$x(t) - 2 = -t + t^3.$$

Thus

$$x(t) = 2 - t + t^3.$$

17.3 Reading the result

The result is not just a cubic polynomial. It tells us how the object moves. Initially, $v(0) = -1$, so it moves in the negative direction. The velocity becomes zero when

$$-1 + 3t^2 = 0,$$

so

$$t = \frac{1}{\sqrt{3}}$$

for $t \geq 0$. At that time the object stops momentarily and then moves in the positive direction, because for larger t the velocity becomes positive.

This interpretation is part of the solution. Without it, we have only algebra.

18 Worked example: two-dimensional parametric motion

Consider

$$\mathbf{r}(t) = (t^2)\mathbf{i} + (2t)\mathbf{j}, \qquad t \geq 0.$$

We want to understand the motion, not just compute derivatives.

18.1 Components

The coordinate functions are

$$x(t) = t^2, \qquad y(t) = 2t.$$

The velocity is

$$\mathbf{v}(t) = 2t\mathbf{i} + 2\mathbf{j}.$$

The acceleration is

$$\mathbf{a}(t) = 2\mathbf{i}.$$

So acceleration is constant and horizontal. The vertical velocity is constant. The horizontal velocity grows linearly.

18.2 Trajectory

To find the geometric path, eliminate time. Since $y = 2t$,

$$t = \frac{y}{2}.$$

Then

$$x = t^2 = \frac{y^2}{4}.$$

Equivalently,

$$y = 2\sqrt{x}$$

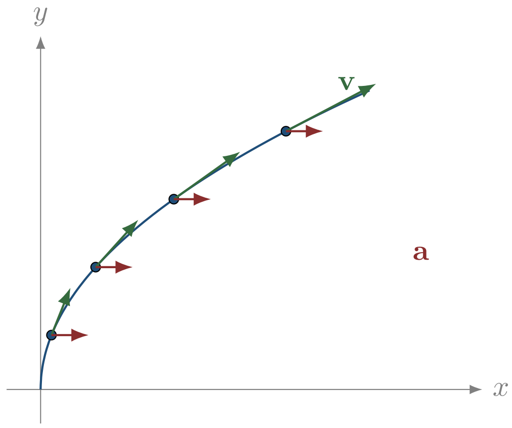
for $y \geq 0$. The trajectory is a sideways parabola.

18.3 Motion along the trajectory

The trajectory alone does not say how fast the object moves. The speed is

$$|\mathbf{v}(t)| = \sqrt{(2t)^2 + 2^2} = 2\sqrt{t^2 + 1}.$$

This increases with time. Therefore, the object moves along the sideways parabola with increasing speed. The acceleration is horizontal, so the velocity vector bends progressively toward the positive x direction.


y
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19 Worked example: interpreting sinusoidal motion

Consider

$$x(t) = A\cos(\omega t).$$

This is a common kinematic form. We should read it carefully.

19.1 Velocity and acceleration

Differentiate:

$$v(t) = -A\omega\sin(\omega t).$$

Differentiate again:

$$a(t) = -A\omega^2\cos(\omega t).$$

Since $x(t) = A\cos(\omega t)$, we can write

$$a(t) = -\omega^2 x(t).$$

This relation says acceleration is always opposite to displacement and proportional to it.

19.2 What the constants mean

The constant A is the amplitude. It determines the maximum displacement from the origin. The constant ω is angular frequency. It determines how quickly the motion repeats. The period is

$$T = \frac{2\pi}{\omega}.$$

This relation is not about circular motion only. It is about any sinusoidal time dependence.

19.3 Phase and initial conditions

A more general sinusoidal motion is

$$x(t) = A\cos(\omega t + \phi).$$

The phase ϕ determines where in the cycle the object is at $t = 0$. Two motions may have the same amplitude and frequency but different phase. They have the same type of motion, but not the same initial condition.

At $t = 0$,

$$x(0) = A\cos\phi,$$

$$v(0) = -A\omega\sin\phi.$$

Thus phase is not abstract decoration. It encodes initial position and initial velocity.

Sinusoidal kinematics

A sinusoidal position function produces a sinusoidal velocity shifted in phase, and an acceleration proportional to $-x$. The derivatives reveal the internal structure of the motion.

20 How to choose coordinates

Many difficulties in kinematics come from poor coordinate choices. The equations do not become false, but they become harder to read. A coordinate system should be chosen to make the motion or the question simple.

20.1 Choosing an origin

The origin is arbitrary, but not meaningless. If an object starts from a natural reference point, it may be convenient to set $x_0 = 0$. If the ground is important, it may be convenient to set $y = 0$ at ground level. If a motion is circular, it may be convenient to place the origin at the centre.

The choice of origin changes coordinate values, but it does not change physical distances or displacements. If every position is shifted by a constant, velocities and accelerations are unchanged because derivatives of constants vanish.

20.2 Choosing axis directions

Axis direction affects signs. If upward is positive, gravitational acceleration is negative. If downward is positive, the same acceleration is positive. Neither choice is inherently correct. The key is consistency.

For one-dimensional vertical motion, we might write

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

if upward is positive. If downward is positive, the signs change. Students sometimes memorize one version and then apply it in a coordinate system where it no longer fits. The cure is to return to meaning: acceleration points downward.

Sign errors

Most sign errors are not algebra errors. They are interpretation errors. Decide the positive direction before writing components, and then keep that decision throughout the calculation.

20.3 Choosing coordinates adapted to the motion

For straight-line motion, one coordinate may be enough. For projectile-like motion, Cartesian components are natural. For circular or radial motion, polar coordinates may be clearer. The coordinate system is a tool. A good tool makes the structure visible.

But changing coordinates does not change the motion. It only changes the language used to describe it.

21 From data points to kinematic ideas

In real situations we may not start with a clean formula. We may have measurements of position at several times. Kinematics still applies. We estimate velocities from differences and accelerations from changes in velocity.

Suppose positions are measured at equal time intervals Δt. The average velocity between two measurements is

$$v_{\text{avg}} \approx \frac{x_{i+1} - x_i}{\Delta t}.$$

A better estimate of instantaneous velocity at the middle point is the central difference

$$v(t_i) \approx \frac{x_{i+1} - x_{i-1}}{2\Delta t}.$$

Acceleration can be estimated by

$$a(t_i) \approx \frac{x_{i+1} - 2x_i + x_{i-1}}{(\Delta t)^2}.$$

These formulas are not separate from calculus. They are finite approximations to derivatives. The smaller the time interval and the smoother the motion, the better they approximate the derivative.

21.1 Why measurement makes interpretation harder

Measured data contain noise. Differentiation tends to amplify noise because it looks at differences between nearby values. This is why experimental velocity and acceleration estimates can be less stable than position measurements. Even in a basic physics course, it is useful to know that real kinematics is not only about exact symbolic functions.

21.2 The conceptual lesson

Whether we have a formula, a graph, or a table of measurements, the same conceptual structure remains:

$$\text{position} \leftrightarrow \text{velocity} \leftrightarrow \text{acceleration}.$$

Calculus gives the exact connection for smooth functions. Finite differences give approximate connections for data.

22 How to read a complete kinematic problem

A kinematic problem should not begin with formula hunting. It should begin with reading. The following procedure is often more useful than any list of equations.

A kinematic reading procedure

1. Identify the object and the time interval.
2. Choose a coordinate system and write what the coordinates mean.
3. List the known functions or values: position, velocity, acceleration, initial conditions.
4. Decide what the question is asking: position, velocity, acceleration, time, displacement, distance, maximum, turning point, or trajectory.
5. Translate the physical event into a mathematical condition, such as $v(t) = 0$, $x(t) = x_{\text{target}}$, or $a(t) = 0$.
6. Differentiate or integrate only when the question requires it.
7. Interpret the result with units, signs, and physical meaning.

22.1 Event conditions

Many questions in kinematics are really event questions. An event is something that happens at a particular time. Examples:

- The object reaches a certain position: $x(t) = x_*$.
- The object stops momentarily: $v(t) = 0$.
- The object reaches maximum height: $v_y(t) = 0$ in a vertical model.
- The object returns to the starting point: $x(t) = x(0)$ or $\mathbf{r}(t) = \mathbf{r}(0)$.
- The speed has a specified value: $|\mathbf{v}(t)| = v_*$.

The most important step is often not solving the equation. It is choosing the correct equation that represents the event.

22.2 Physical domains

Mathematical equations may have solutions that are not physically relevant. If time starts at $t = 0$, we usually require

$$t \geq 0.$$

If a square root appears, the expression under the root must be non-negative. If a model applies only before collision or before landing, solutions outside that interval should not be used.

Mathematical solution versus physical answer

Solving an equation is not the end. You must check whether the solution belongs to the physical domain of the problem.

23 Distance traveled versus displacement

Displacement and distance traveled are often confused. Displacement is the change in position:

$$\Delta x = x(t_2) - x(t_1).$$

Distance traveled is the total length of the path traveled. In one dimension, it is

$$\int_{t_1}^{t_2} |v(t)| \, \mathrm{d}t.$$

The absolute value is essential. If the object moves forward and then backward, displacements can cancel, but distance traveled does not cancel.

23.1 Example

Let

$$x(t) = t^2 - 2t, \qquad 0 \le t \le 3.$$

Then

$$v(t) = 2t - 2.$$

The velocity is zero at $t = 1$. The object moves in the negative direction for $0 < t < 1$ and in the positive direction for $t > 1$.

The displacement over the interval is

$$x(3) - x(0) = (9 - 6) - 0 = 3.$$

But the distance traveled is not 3. We split the interval at the turning point:

$$x(0) = 0, \qquad x(1) = -1, \qquad x(3) = 3.$$

From $t = 0$ to $t = 1$, the object travels distance 1. From $t = 1$ to $t = 3$, it travels distance 4. Total distance is

$$1 + 4 = 5.$$

This example shows why the sign of velocity matters. Displacement is a vector-like signed change; distance is accumulated path length.

24 Maxima, minima, and turning points

Calculus is often used in kinematics to identify special moments. A maximum or minimum of position occurs when the derivative is zero, provided the point is inside the time interval and the derivative changes sign appropriately.

In one-dimensional motion, a turning point occurs when

$$v(t) = 0$$

and the object changes direction. But not every zero of velocity is necessarily a turning point in a broader mathematical sense. The interpretation depends on what happens before and after.

24.1 Maximum height as a kinematic condition

In vertical motion, maximum height occurs when vertical velocity is zero:

$$v_y(t) = 0.$$

This condition is not a memorized rule. It comes from the fact that height is a function of time, and at an interior maximum its derivative is zero.

If

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2,$$

then

$$v_y(t) = v_0 - gt.$$

Set

$$v_0 - gt = 0,$$

so

$$t = \frac{v_0}{g}.$$

This time is meaningful only if $v_0 > 0$ in a coordinate system where upward is positive. If $v_0 < 0$, the object is initially moving downward and there is no later upward maximum after launch.

24.2 Maximum speed?

If a problem asks for maximum speed, the condition is different. Speed is

$$|\mathbf{v}(t)|.$$

A maximum of speed may require differentiating the speed or the square of speed:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(|\mathbf{v}(t)|^2\right) = 0.$$

This illustrates a broader lesson: different physical quantities require different mathematical conditions. Do not automatically set $v = 0$ whenever you see the word maximum.

Translate before solving

Maximum position, maximum height, maximum speed, and maximum distance from the origin are different questions. Each has its own mathematical expression and condition.

25 Relative motion

Kinematics also describes motion relative to another moving object. Suppose two particles have positions $\mathbf{r}_A(t)$ and $\mathbf{r}_B(t)$. The position of A relative to B is

$$\mathbf{r}_{A/B}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t).$$

The relative velocity is

$$\mathbf{v}_{A/B}(t) = \mathbf{v}_A(t) - \mathbf{v}_B(t),$$

and the relative acceleration is

$$\mathbf{a}_{A/B}(t) = \mathbf{a}_A(t) - \mathbf{a}_B(t).$$

25.1 Meaning of relative position

If $\mathbf{r}_{A/B} = 0$, then the objects are at the same position. If $\mathbf{r}_{A/B}$ is constant, then A stays fixed relative to B. If $\mathbf{v}_{A/B} = 0$, then they have the same velocity at that moment.

This language is useful in pursuit problems, collision predictions, and motion viewed from moving platforms. Again, the mathematical operation is simple subtraction, but the interpretation is physical.

25.2 Example

Let

$$x_A(t) = 2 + 3t, \qquad x_B(t) = 10 - t.$$

Then

$$x_{A/B}(t) = x_A(t) - x_B(t) = 2 + 3t - (10 - t) = -8 + 4t.$$

They meet when

$$x_{A/B}(t) = 0,$$

so

$$-8 + 4t = 0, \qquad t = 2.$$

The relative velocity is

$$v_{A/B} = 4,$$

meaning that A closes the gap relative to B at four units of length per unit time.

26 Motion constrained to a path

Sometimes the path is known in advance. For example, a train moves along a track, a bead moves along a wire, or a point moves along a prescribed curve. Then it may be useful to describe position by a path coordinate $s(t)$, the distance measured along the curve.

If $s(t)$ is the arc length along the path, then speed is

$$v(t) = \frac{\mathrm{d}s}{\mathrm{d}t}.$$

Acceleration along the path is

$$a_t(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2}.$$

But if the path is curved, there is also normal acceleration

$$a_n = \frac{v^2}{\rho},$$

where ρ is the radius of curvature.

26.1 Why this is still kinematics

No force has been mentioned. We are only describing what acceleration must be present if the particle follows a curved path with a certain speed. Dynamics would ask what physical interaction supplies that acceleration. Kinematics asks what acceleration is required by the geometry and timing of the motion.

26.2 A small correction about notation

The path coordinate s should not be confused with time. In many integrals, one uses a dummy variable such as s for time integration. Here s means arc length. Notation is a tool, but it must be read carefully from context.

27 Piecewise motion

Many real motions have different phases. A car may accelerate, then move at constant speed, then brake. The equation of motion is then piecewise.

For example, suppose

$$a(t) = \begin{cases} 2, & 0 \le t < 3, \\ 0, & 3 \le t < 7, \\ -4, & 7 \le t \le 9. \end{cases}$$

To build $v(t)$ and $x(t)$, we integrate separately on each interval and enforce continuity of position and velocity at the transition times.

27.1 Why continuity matters

If position suddenly jumped, the object would teleport. In ordinary mechanics, position is continuous. Velocity may change rapidly, but an instantaneous jump in velocity would require an impulsive acceleration. In elementary piecewise models, we usually keep velocity continuous unless the model explicitly includes an impulse.

27.2 Constructing the first interval

Assume $x(0) = 0$ and $v(0) = 0$. For $0 \le t < 3$,

$$v(t) = 2t,$$

$$x(t) = t^2.$$

At $t = 3$,

$$v(3) = 6, \qquad x(3) = 9.$$

These become the initial data for the next interval.

27.3 Second interval

For $3 \le t < 7$, acceleration is zero, so velocity remains 6. Using $t = 3$ as the local initial moment,

$$x(t) = 9 + 6(t - 3).$$

At $t = 7$,

$$x(7) = 9 + 24 = 33, \qquad v(7) = 6.$$

The third interval would start from these data.

The main lesson is that each phase has its own equation, but the phases are connected by physical continuity and by the end data of the previous phase.

28 Mistakes that reveal misunderstanding

Errors in kinematics are often not random. They reveal specific conceptual confusions. Recognizing these confusions is useful because it tells us what to repair.

28.1 Mistake 1: confusing position and displacement

Position is measured relative to an origin. Displacement is a change in position. If an object moves from $x = 5$ to $x = 8$, its displacement is 3, not 8.

28.2 Mistake 2: confusing velocity and speed

Velocity has direction. Speed does not. In one dimension, velocity can be negative; speed cannot.

28.3 Mistake 3: assuming acceleration means speeding up

Acceleration means change of velocity. It may slow an object down, speed it up, or change only its direction.

28.4 Mistake 4: using constant-acceleration formulas when acceleration is not constant

The equation

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

requires constant acceleration. If $a = a(t)$, this expression is generally wrong. You must integrate the actual acceleration function.

28.5 Mistake 5: ignoring initial conditions

A derivative equation without initial conditions usually describes a family of motions. Constants of integration must be determined from physical data.

28.6 Mistake 6: treating trajectory as motion

The curve $y = x^2$ is a path. It does not tell how quickly the object moves along it. Parametric equations with time describe motion.

28.7 Mistake 7: forgetting the physical domain

A quadratic equation may produce two times, but only one may satisfy $t \geq 0$ or belong to the interval before collision or landing. Mathematical roots must be filtered by physical meaning.

29 A compact dictionary of kinematic translations

The table below summarizes common physical statements and their mathematical translations. It is not meant to replace understanding. It is meant to show how verbal descriptions become equations.

Physical statement	Mathematical translation	Typical operation
The object is at a given position	$x(t) = x_*$ or $\mathbf{r}(t) = \mathbf{r}_*$	solve for t
The object stops momentarily	$v(t) = 0$	solve for time
The speed has a given value	$\lvert\mathbf{v}(t)\rvert = v_*$	solve equation
The acceleration is zero	$a(t) = 0$	solve or interpret
The object reaches maximum position	$v(t) = 0$ plus sign/interval check	derivative test
Displacement over an interval	$x(t_2) - x(t_1)$	evaluate position
Distance traveled	$\int_{t_1}^{t_2} \lvert v(t)\rvert \, \mathrm{d}t$	integrate speed
Velocity from position	$v = x'$	differentiate
Acceleration from velocity	$a = v'$	differentiate
Velocity from acceleration	$v = v_0 + \int a \, \mathrm{d}t$	integrate
Position from velocity	$x = x_0 + \int v \, \mathrm{d}t$	integrate
Trajectory from parametric motion	eliminate t	algebra

This dictionary is useful only if each entry is read with meaning. For example, $v(t) = 0$ is not a universal answer to every question involving a maximum. It is the correct condition for an interior maximum or minimum of position in one-dimensional motion, and for a momentary stop. For maximum speed, the condition is different.

30 Guided problems with full reading

30.1 Problem 1: from position to motion story

Let

$$x(t) = 3 + 6t - 2t^2, \qquad t \geq 0.$$

Describe the motion, find when the object stops, and find its position at that moment.

Reading the function

The function is quadratic, so the acceleration is constant. Differentiate:

$$v(t) = 6 - 4t,$$

$$a(t) = -4.$$

At $t = 0$, the object is at $x = 3$ and has velocity 6. Since acceleration is negative, velocity decreases.

Stopping condition

The object stops momentarily when

$$v(t) = 0.$$

Thus

$$6 - 4t = 0, \qquad t = \frac{3}{2}.$$

The position is

$$x\left(\frac{3}{2}\right) = 3 + 6 \cdot \frac{3}{2} - 2\left(\frac{3}{2}\right)^2 = 3 + 9 - \frac{9}{2} = \frac{15}{2}.$$

So the object stops at $x = 7.5$ and then moves in the negative direction.

30.2 Problem 2: from acceleration to equation of motion

An object has acceleration

$$a(t) = 4 - 2t,$$

with $x(0) = 1$ and $v(0) = 3$. Find $x(t)$.

First integrate acceleration:

$$v(t) = 3 + \int_0^t (4 - 2s)\,\mathrm{d}s = 3 + 4t - t^2.$$

Then integrate velocity:

$$x(t) = 1 + \int_0^t (3 + 4s - s^2)\,\mathrm{d}s = 1 + 3t + 2t^2 - \frac{1}{3}t^3.$$

The initial conditions are visible in the final expression: $x(0) = 1$ and $v(0) = 3$.

30.3 Problem 3: trajectory and velocity vectors

Let

$$\mathbf{r}(t) = t\mathbf{i} + (1 - t^2)\mathbf{j}.$$

Then

$$\mathbf{v}(t) = \mathbf{i} - 2t\mathbf{j},$$

and

$$\mathbf{a}(t) = -2\mathbf{j}.$$

Eliminating time gives $x = t$, so

$$y = 1 - x^2.$$

The path is a parabola, but the velocity information comes from the parametric form. At $t = 0$, velocity is horizontal. At later positive times, the vertical component is negative, so the object moves downward along the parabola.

31 A longer example: designing a motion from requirements

Kinematics is not only used to analyze a given motion. It can also be used to construct a motion satisfying requirements. Suppose we want one-dimensional motion with the following properties:

- at $t = 0$, the object is at $x = 0$;
- at $t = 0$, the object is at rest;
- at $t = T$, the object reaches $x = L$;
- at $t = T$, the object is again at rest.

We want a simple polynomial motion $x(t)$ satisfying these conditions.

A linear function cannot work because it would have constant velocity and could not be at rest at both endpoints unless it never moved. A quadratic function has three coefficients, but we have four conditions. So we try a cubic:

$$x(t) = at^3 + bt^2 + ct + d.$$

Apply the initial conditions. From $x(0) = 0$, we get

$$d = 0.$$

The velocity is

$$v(t) = 3at^2 + 2bt + c.$$

From $v(0) = 0$, we get

$$c = 0.$$

So

$$x(t) = at^3 + bt^2.$$

Now impose the final conditions:

$$x(T) = aT^3 + bT^2 = L,$$

$$v(T) = 3aT^2 + 2bT = 0.$$

From the velocity condition,

$$b = -\frac{3}{2}aT.$$

Substitute into the position condition:

$$aT^3 - \frac{3}{2}aT^3 = L,$$

so

$$-\frac{1}{2}aT^3 = L, \qquad a = -\frac{2L}{T^3}.$$

Then

$$b = \frac{3L}{T^2}.$$

The designed motion is

$$x(t) = 3L\left(\frac{t}{T}\right)^2 - 2L\left(\frac{t}{T}\right)^3.$$

This example is important because it reverses the usual classroom habit. We did not start with a formula. We started with desired physical behavior and constructed a function that has that behavior.

32 Kinematic thinking in algorithmic form

Although these notes are not about numerical methods, it is useful to summarize kinematic reasoning as a sequence of decisions. This is not because physics is programming, but because a precise sequence helps prevent formula hunting.

1. Decide what the unknown function is: $x(t)$, $\mathbf{r}(t)$, $v(t)$, or $a(t)$.
2. Decide what information is directly given.
3. If position is given and velocity or acceleration is needed, differentiate.
4. If acceleration or velocity is given and position is needed, integrate.
5. Insert initial conditions at the correct stage.
6. Translate events into equations.
7. Solve the equations in the physically allowed domain.
8. Interpret signs, units, and directions.

This sequence is simple, but it is powerful. It keeps the solution anchored in meaning.

32.1 A final contrast

A weak solution says:

> I used the formula and got the answer.

A strong solution says:

> The object reaches the required point when its position function satisfies this condition. I used the given acceleration to reconstruct velocity and position, applied the initial conditions, solved the event equation, and checked that the time is physically admissible.

The second solution is longer, but it reveals understanding. In physics, understanding is not the enemy of calculation. It is what makes calculation meaningful.

33 One-page summary

Core relations

$$\mathbf{v}(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}, \qquad \mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}.$$

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(s)\,\mathrm{d}s, \qquad \mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(s)\,\mathrm{d}s.$$

Constant acceleration in one dimension

$$v(t) = v_0 + at,$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2,$$

$$v^2 = v_0^2 + 2a(x - x_0).$$

These relations require constant acceleration.

Graphs

- Slope of $x(t)$ is velocity.
- Slope of $v(t)$ is acceleration.
- Area under $v(t)$ is displacement.
- Area under $a(t)$ is change in velocity.

Interpretation rules

- Velocity is tangent to the trajectory.
- Speed is the magnitude of velocity.
- Acceleration means change of velocity, not necessarily increase of speed.
- Initial conditions select one motion from a family of possible motions.
- A trajectory is a path; a motion is a path with timing.

34 Additional practice questions

These questions are designed to train translation between words, equations, graphs, and interpretation. Do not begin by searching for a formula. Begin by identifying what is given and what condition answers the question.

1. Given $x(t) = 5 - 4t + t^2$, find $v(t)$, $a(t)$, the stopping time, and the position at the stopping time.
2. Given $a(t) = 2t + 1$, $v(0) = 0$, and $x(0) = 3$, find $v(t)$ and $x(t)$.
3. Given $\mathbf{r}(t) = (2t)\mathbf{i} + (3 - t^2)\mathbf{j}$, find velocity, acceleration, speed, and the trajectory equation.
4. Construct a motion with constant acceleration that starts from $x = 1$ with velocity -2 and has acceleration 3.
5. Find the displacement and distance traveled for $x(t) = t^2 - 4t$ on $0 \le t \le 5$.
6. For $x(t) = A\sin(\omega t)$, find $v(t)$, $a(t)$, and the relation between acceleration and position.
7. A velocity graph is $v(t) = 3 - 2t$. Find displacement from $t = 0$ to $t = 4$ and distance traveled over the same interval.
8. A particle has $\mathbf{v}(t) = t\mathbf{i} + t^2\mathbf{j}$ and $\mathbf{r}(0) = \mathbf{0}$. Find $\mathbf{r}(t)$.
9. Explain in words why an object can have acceleration even if its speed is constant.
10. Give an example of two different motions with the same trajectory but different timing.

Closing perspective

Kinematics is sometimes treated as the easy beginning of mechanics, a short chapter before the real physics starts. That is a mistake. Kinematics teaches the language in which the rest of mechanics is written. If a student does not understand what position, velocity, acceleration, derivative, integral, graph slope, graph area, trajectory, and initial condition mean, then later formulas in dynamics will look like disconnected recipes.

The goal is therefore not to memorize many equations. The goal is to learn how equations describe motion. When we know position, differentiation tells us velocity and acceleration. When we know acceleration, integration and initial conditions reconstruct velocity and position. When we draw a graph, slopes and areas become physical quantities. When we draw a trajectory, tangent vectors and acceleration vectors reveal the geometry of motion. When we solve an event equation, we translate a physical sentence into a mathematical condition.

This is the habit that should remain after the details are forgotten:

Do not ask first: which formula? Ask first: what is changing, and how is that change encoded mathematically?

If that question becomes natural, kinematics stops being a list of formulas and becomes a working language for mechanics.